
Supplementary information

scSAGE: Integrating Self-Expression and Graph Contrastive Learning for Comprehensive Analysis of Single-Cell RNA-seq Data

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Supplementary Text

A. Proofs of Lemma 1 and Theorem 1

Lemma 1. *According to the objective function in Eq.(20) in the main text, we first take the derivative of Eq.(20) in the main text and set it equal to zero to obtain the closed-form solution for C .*

$$C^* = ZZ^T(ZZ^T + \mathbf{I})^{-1}, \quad (1)$$

Proof. According to the objective function in Eq.(20) in the main text, we first calculate the closed-form solution for C .

$$\begin{aligned} \mathcal{L}(C) &= \|Z - CZ\|_F^2 + \|C\|_2^2 \\ &= \text{Tr}((Z - CZ)^T(Z - CZ)) + \text{Tr}(C^T C) \end{aligned} \quad (2)$$

where Tr represents the trace of the matrix. Then we have

$$\frac{\alpha \mathcal{L}}{\alpha C} = -2ZZ^T + 2CZZ^T + 2C \quad (3)$$

Let Eq.(3) equal to 0, we can obtain the closed-form solution C^* of Eq.(20) in the main text, as shown in Eq.(1). QED

Based on Lemma 1, we can demonstrate that the self-expressive matrix C can capture similarities in the original gene expression matrix through Theorem 1.

Theorem 1. *The self-expressive matrix $C \in \mathbb{R}^{N \times N}$ and the gene expression value x_i and x_j of any two cells have the same similarity, when the condition $x_i \rightarrow x_j$ holds, i.e.*

$$\begin{aligned} x_i \rightarrow x_j &\Rightarrow |c_{ip} - c_{jp}| \rightarrow 0 \\ &\forall i, j, p : 1 \leq i, j, p \leq N, \end{aligned} \quad (4)$$

Proof. For $1 \leq i \leq N$, we denote c_i as the i -th row of the coefficient matrix C . According to Eq.(20) in the main text, we let $\mathcal{L}(c_i) = \|Z - c_i Z\|_F^2 + \|c_i\|_2^2$. Similar to Eq.(3), we differentiate the above equation and set it equal to 0, as illustrated below:

$$\begin{aligned} \frac{\alpha \mathcal{L}}{\alpha c_{ip}} &= -(z_i - c_i^* Z)z_p^T + c_{ip} = 0 \\ c_{ip} &= (z_i - c_i^* Z)z_p^T \end{aligned} \quad (5)$$

Based on Eq.(5), we can obtain:

$$\begin{aligned} c_{ip} - c_{jp} &= (z_i - c_i^* Z)z_p^T - (z_j - c_j^* Z)z_p^T \\ &= (z_i - z_j)z_p^T - (c_i^* - c_j^*)Zz_p^T \end{aligned} \quad (6)$$

Moreover, according to Eq.(1), we have

$$\begin{aligned} c_i^* - c_j^* &= z_i Z^T (ZZ^T + \mathbf{I}_N)^{-1} - z_j Z^T (ZZ^T + \mathbf{I}_N)^{-1} \\ &= (z_i - z_j)Z^T (ZZ^T + \mathbf{I}_N)^{-1} \end{aligned} \quad (7)$$

Substitute $c_i^* - c_j^*$ into $c_{ip} - c_{jp}$.

$$c_{ip} - c_{jp} = (z_i - z_j)z_p^T - (z_i - z_j)Z^T (ZZ^T + \mathbf{I}_N)^{-1} Z z_p^T \quad (8)$$

Replace $(ZZ^T + \mathbf{I}_N)^{-1} Z z_p^T$ with M

$$c_{ip} - c_{jp} = (z_i - z_j)z_p^T - (z_i - z_j)Z^T M \quad (9)$$

Therefore, we have

$$|c_{ip} - c_{jp}| \leq \|z_i - z_j\|_2 \|z_p^T - Z^T M\|_2 \quad (10)$$

If the condition that $x_i \rightarrow x_j$ holds, i.e., $\|x_i - x_j\|_2 \rightarrow 0$ hold, we can obtain $\|z_i - z_j\|_2 \rightarrow 0$ according to Eq.(3) in the main text. As a result, Eq.(10) close to 0, and $\|c_{ip}^* - c_{jp}^*\| \rightarrow 0$ holds, thus C has the clustering effect. QED

B. Implementation of the comparison methods

In this study, we evaluated scSAGE by comparing it with several state-of-the-art methods, including scTAG [1], scDEFR [2], scGCOT [3], scDFC [4], scNAME [5], contrastive-sc (*abbr.con-sc*) [6], scCCL [7], scGCL [8], k-means [9] and spectral clustering [10].

- ZINB-Based Graph Embedding Autoencoder (scTAG): The proposed method is a single-cell clustering approach based on deep graph embedding. It combines the ZINB model with a topologically adaptive graph convolutional autoencoder.
- Unsupervised Gene-Cell collective representation learning and Optimal Transport (scGCOT): A novel end-to-end deep graph clustering model for single-cell transcriptomics data based on unsupervised gene-cell collective representation learning and optimal transport which integrates both cell and gene correlations.
- Single-cell graph neural network (scGNN): scGNN is an algorithm that utilizes graph neural networks to learn the relationships and feature representations between cells for single-cell clustering.
- Deep fusion clustering method (scDFC): scDFC is a single-cell deep fusion clustering model that comprises an attributed feature clustering module and a structure-attention feature clustering module. It integrates attribute, structural, and attentional information on scRNA-seq data.
- Unsupervised Deep Embedded Fusion Representation (scDEFR): Using a model based on deep fusion embedding representation, fusion embedding is generated to learn a merged representation that includes both gene-level transcriptomic and cellular topological information.
- Contrastive self-supervised clustering (contrastive-sc): An unsupervised deep learning method is utilized to learn low-dimensional representations of scRNA-seq data using contrastive representation learning. Subsequently, a general clustering algorithm is applied to cluster the data.
- Neighborhood contrastive with ancillary mask estimation (scNAME): This algorithm integrates a mask estimation task for mining gene correlations, a neighborhood contrastive learning framework for developing intrinsic cell structures, and a global memory bank. This combination enables the model to capture discriminative representations globally.
- Self-Supervised Contrastive Learning (scCCL): scCCL is a novel self-supervised contrastive learning method that applies contrastive learning separately to instance-level contrastive learning module and cluster-level contrastive learning mode.
- Imputation method based on graph contrastive learning (scGCL): A single-cell graph contrastive learning method, it is proposed for scRNA-seq data imputation. This method integrates graph contrastive learning and zero-inflated negative binomial distribution to estimate missing values.

C. Evaluation metrics

We use adjusted rand index (ARI) [11] and normalized mutual information (NMI) [12] to assess clustering results.

ARI. It measures the agreement between two clusterings by considering pairwise sample assignments. ARI ranges from -1 to 1, with higher values indicating better agreement. In the equation (11), N represents the

number of samples, $\mathcal{C}^{d'}$ represents the sample set of the d' -th predicted cluster, $\mathcal{G}^d$ represents the samples set of the d -th ground truth cluster. $E[RI]$ is the expected random index.

$$E[RI] = [\sum_{d'} \binom{|\mathcal{C}^{d'}|}{2} \sum_d \binom{|\mathcal{G}^d|}{2}] / \binom{N}{2}$$

$$ARI = \frac{\sum_{d,d'} (|\mathcal{G}^d \cap \mathcal{C}^{d'}|) - E[RI]}{\frac{1}{2}[\sum_d \binom{|\mathcal{C}^d|}{2} + \sum_j \binom{|\mathcal{G}^j|}{2}] - E[RI]} \quad (11)$$

NMI. It leverages information theory to quantify the similarity between clusterings. NMI ranges from 0 to 1, with higher values indicating better agreement between the clusterings. In the equation (12), N represents the number of samples, $I(\mathcal{C}; \mathcal{G})$ denotes the mutual information between the predicted clusters $\mathcal{C}$ and the ground truth clusters $\mathcal{G}$, and $H(\mathcal{G})$ denotes the information entropy of the $\mathcal{G}$.

$$NMI = \frac{2 \cdot I(\mathcal{C}; \mathcal{G})}{H(\mathcal{C}) + H(\mathcal{G})}$$

$$I(\mathcal{C}; \mathcal{G}) = \sum_{d'} \sum_d P(V^{d'}, \mathcal{G}^d) \log \frac{P(\mathcal{C}^{d'}, \mathcal{G}^d)}{P(\mathcal{C}^{d'})P(\mathcal{G}^d)}$$

$$H(\mathcal{G}) = - \sum_d P(\mathcal{G}^d) \log P(\mathcal{G}^d)$$

$$P(\mathcal{G}^{d'}, \mathcal{C}^d) = \frac{|\mathcal{G}^{d'} \cap \mathcal{C}^d|}{N}, \quad P(\mathcal{G}^d) = \frac{|\mathcal{G}^d|}{N} \quad (12)$$

D. Confidence level comparison

To further validate the superiority of our approach, we calculated the 95% confidence intervals for the ARI and NMI results of our method and the most competitive method in the baseline as Table.S1. The results indicate that at the 95% confidence level, our method's ARI and NMI performance fall within [0.85677, 0.95657] and [0.79164, 0.94220], respectively. The narrower width of these intervals and their higher lower bounds suggest more precise and stable performance estimates, indicating superior average performance compared to other competitive methods. This also statistically validates the superiority of our method.

Table S1. 95% confidence interval comparison.

Method	ARI	NMI
Ours	[0.85677, 0.95657]	[0.79164, 0.94220]
scDEFR	[0.76455, 0.97723]	[0.76455, 0.97723]
scNAME	[0.76589, 0.93227]	[0.67311, 0.94639]
scCCL	[0.76979, 0.91351]	[0.69963, 0.93107]
scGCL	[0.75578, 0.89474]	[0.62155, 0.88787]

E. Complexity analysis

Based on the algorithm proposed in this research, we calculated the following time complexity. We use N to represent the number of cells, D to denote the quantity of highly variable genes, d to represent the dimensionality of embeddings, and K to indicate the number of categories. In the objective function Eq.(10) in the main text, where $C_{p[i,:]}^{v_1} \in \mathbb{R}^d$, the time complexity can be calculated as $O(K \cdot N \cdot d)$. In the objective function Eq.(11) in the main text, where $CEN_p^{v_1} \in \mathbb{R}^d$, the time complexity can be calculated as $O(K \cdot d^2)$. In Eq.(17) in the main text of the objective function, the time complexity is predominantly influenced by Eq.(15) in the main text. Consequently, the time complexity can be computed as $O(N \cdot D)$. In the objective function Eq.(18) in the main text, where $\tilde{X} \in \mathbb{R}^{N \times D}$, the time complexity can be calculated as $O(N \cdot D)$.

In the objective function Eq.(20) in the main text, where $\tilde{Z} \in \mathbb{R}^{N \times d}$ and $\tilde{C} \in \mathbb{R}^{N \times N}$, the time complexity can be calculated as $O(N^2 \cdot d)$.

In conclusion, the overall time complexity of the algorithm can be calculated as follows:

$$\begin{aligned} &O(K \cdot N \cdot d + K \cdot d^2 + N \cdot D + N \cdot D + N^2 \cdot d) \\ &= O(K \cdot N \cdot d + K \cdot d^2 + N \cdot D + N^2 \cdot d) \end{aligned} \quad (13)$$

Since N is much larger than d and K , the time complexity can be simplified as follows.

$$O(N + K \cdot d^2 + N \cdot D + N^2) \quad (14)$$

F. Implementation Details

Our experimental setup includes a desktop computer with an Intel Core i5-10500 CPU, an NVIDIA GeForce RTX 3090Ti GPU, 16.0 GB RAM, and the PyTorch platform. In the KNN algorithm used to construct the cell graph, we set the parameter k representing the number of nearest neighbors, to 10. For the initial two MLPs with non-shared parameters, their structure is configured as a two-layer MLP network with a hidden layer consisting of 250 nodes. For the total loss Eq.(22) in the main text that needs to be minimized, we employ the Adam [13] optimization algorithm. The learning rate is set to 0.002, and the maximum number of training epochs is set to 500. For the baselines, the parameter settings are the same as those in the original model. We train all models 10 times and average the results for comparison.

G. Hyperparameter sensitivity analysis

Impact of Parameter k .

The parameter k plays a crucial role in constructing the cell graph using KNN. To examine the impact of this hyperparameter on scSAGE, we conduct experiments with k values ranging from 5 to 30. Figure S1(A) and Supplementary Table S5 displays the average ARI and NMI values obtained across all datasets for different k values. This figure clearly shows that the model's performance reaches its peak when $k=10$. As a result, we set the parameter k to 10 in this paper.

Different numbers of highly variable genes.

The selection of HVGs is crucial for single-cell analysis, as it can help emphasize the heterogeneity between cells and enable more accurate clustering. To explore the influence of the number of selected HVGs on the model performance, we utilize 300, 500, 1000, 1500, and 2000 HVGs as inputs for the model across all datasets. The model's performance is depicted in Figure S1(B) and Supplementary Table S6. As shown in this figure, the model achieves the best performance when utilizing 2000 HVGs as inputs.

Sensitivity Analysis of Noise λ .

The hyperparameter λ controls the degree of Gaussian noise added to ensure that the parameters of the two Siamese encoders in the model are not shared. This prevents the generation of shortcut paths in the Siamese encoders, preserving the performance of subsequent contrastive learning. In this section, we conduct an in-depth analysis of the hyperparameter λ , compare the performance of the models with λ ranging from 0.01 to 0.09. The results are presented in the Figure S1(C) and Supplementary Table S7., where we observe that the model's performance reaches its peak when λ is set to 0.07.

Sensitivity Analysis of the Threshold Hyperparameter τ .

Finally, we examine the impact of the threshold hyperparameter τ . As depicted in Figure S1(D) and Supplementary Table S8, we observe that the model achieves the optimal performance when $\tau=0.8$. Moreover, we believe that when $\tau>0.9$, excluding too many samples may result in a small number of positive samples, thereby affecting the discriminative performance of the model. Conversely, when $\tau<0.7$, retaining too many irrelevant samples may introduce bias to the model learning process.

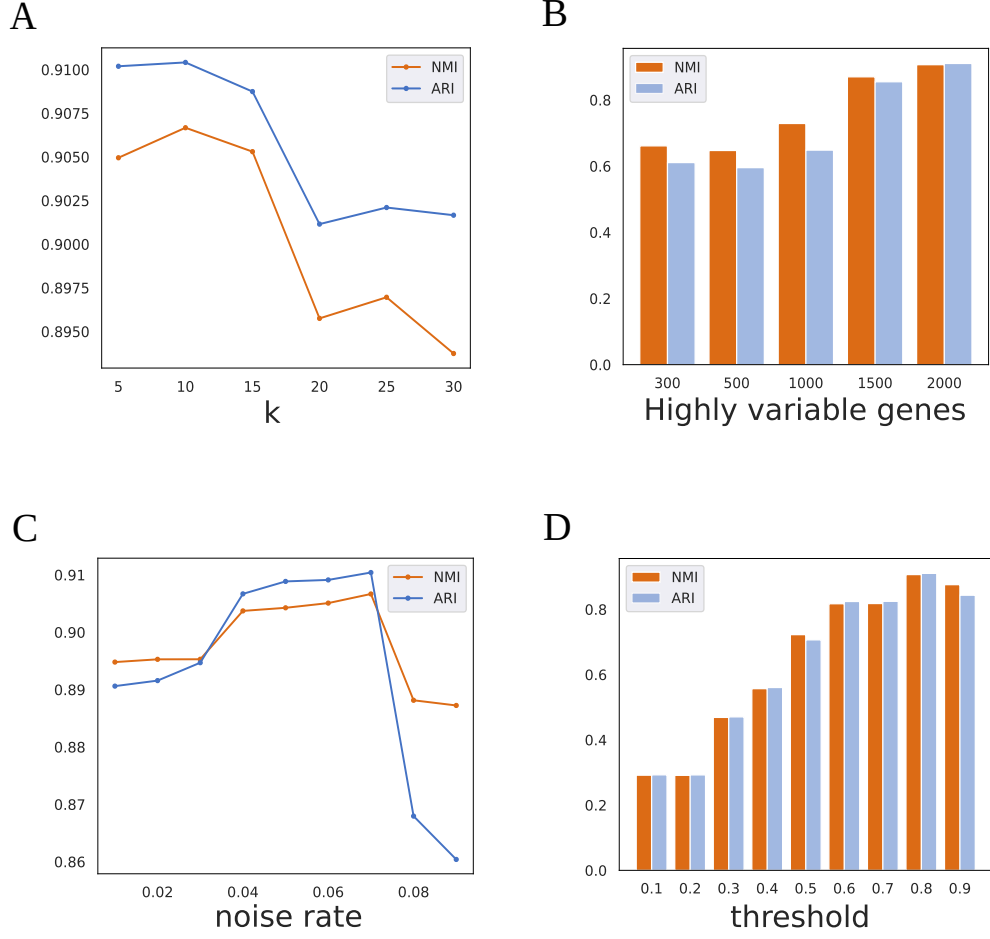


Figure S1. The analysis of the average NMI and ARI with different (A) neighbor parameters k , (B) number of highly variable genes, (C) proportion of noise added to the Siamese encoders and (D) scale threshold of the sample τ .

H. Ablation experiments

For the ablation study, we design four model variants to verify whether each module plays a role in scSAGE. (I) In the scSAGE_{NREC} variant, we remove the reconstruction loss from the model. (II) In the scSAGE_{NSE} variant, we exclude the self-expression module. Instead, we directly apply clustering to the embedding. (III) In the scSAGE_{NCON} variant, we completely remove the CL strategy from the encoder. (IV) In the scSAGE_{NZINB} variant, we remove the ZINB loss from the model.

The results of the ablation experiments are presented in Figure S2(A), Figure S2(B) and Supplementary Table S9. These results clearly indicate that removing any component of scSAGE leads to decreases in the NMI and ARI metrics. Specifically, removing the reconstruction module of the decoder significantly decreases the performance of the model, indicating that the reconstruction loss based on the original gene expression matrix can effectively complement the ZINB loss. Furthermore, we observe that the incorporation of the selfexpression module significantly enhances the performance of our method. This further confirms that the self-expression module reveals intrinsic similarities in single-cell data, enhancing the suitability of learned embeddings for clustering.

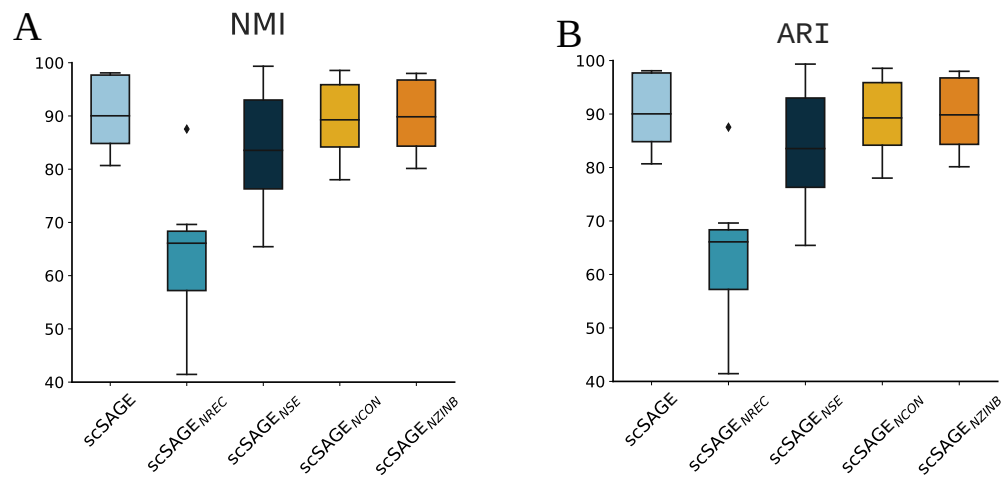


Figure S2. Ablation experiment results: (A) NMI and (B) ARI.

Supplementary Tables

Table S2. The NMI and ARI of scSAGE and its three variants of contrastive learning modules on eight scRNA-seq benchmark datasets. Bold: champion.

	Datasets	scSAGE	scSAGE _{PCA}	scSAGE _{LSTM}
NMI	Qs_Diaphragm	0.9773	0.9601	0.9544
	Qs_Limb_Muscle	0.9753	0.8568	0.874
	Qs_Lung	0.8342	0.8264	0.7406
	Muraro	0.8907	0.8761	0.8323
	Qx_Limb_Muscle	0.9809	0.9798	0.9719
	Young	0.8069	0.6678	0.6471
	Qx_Trachea	0.9012	0.7393	0.7395
	Qs_Trachea	0.8994	0.8062	0.6808
	Qx_Bladder	0.9809	0.7833	0.6533
	Klein	0.8199	0.7997	0.8101
ARI	Qs_Diaphragm	0.9893	0.9778	0.9749
	Qs_Limb_Muscle	0.9875	0.8851	0.8992
	Qs_Lung	0.7972	0.7632	0.4978
	Muraro	0.9282	0.9134	0.8759
	Qx_Limb_Muscle	0.9929	0.9908	0.9878
	Young	0.6929	0.4747	0.4587
	Qx_Trachea	0.9688	0.5746	0.5333
	Qs_Trachea	0.9368	0.8419	0.5538
	Qx_Bladder	0.9893	0.7394	0.5407
	Klein	0.8212	0.7977	0.8164

Table S3. The NMI and ARI of scSAGE and its three variants of contrastive learning modules on eight scRNA-seq benchmark datasets. Bold: champion.

methods	NMI	ARI
scSAGE	0.6657	0.7396
scTAG	0.3118	0.262
scDEFER	0.5422	0.3895
scGCOT	0.172	0.1013
scDFC	—	—
scNAME	0.4019	0.2037
con-sc	0.5135	0.2855
scCCL	0.4653	0.2322
scGCL	0.5317	0.3145
scSAGE _{LSTM}	0.477	0.2384
scSAGE _{PCA}	0.4274	0.1942

Table S4. The average NMI and ARI analysis under different neighborhood parameters k .

	5	10	15	20	25	30
NMI	0.9049	0.9067	0.9053	0.8957	0.8969	0.8937
ARI	0.9102	0.9104	0.9087	0.9012	0.9021	0.9017

Table S5. The average NMI and ARI analysis under different number of highly variable genes.

	300	500	1000	1500	2000
NMI	0.6613	0.6474	0.7289	0.87001	0.9067
ARI	0.6108	0.5955	0.6485	0.8551	0.9104

Table S6. The average NMI and ARI analysis under different proportion of noise added to the Siamese encoders.

	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
NMI	0.8978	0.8952	0.8912	0.9037	0.9042	0.9050	0.9066	0.8881	0.8872
ARI	0.9006	0.8915	0.8763	0.9066	0.9088	0.9091	0.9104	0.8679	0.8604

Table S7. The average NMI and ARI analysis under different scale threshold of the sample τ .

	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
NMI	0.2917	0.2912	0.4688	0.5569	0.7225	0.8172	0.8179	0.9066	0.8757
ARI	0.2927	0.2926	0.4705	0.5605	0.7065	0.8241	0.8246	0.9104	0.8434

Table S8. Ablation studies on scSAGE.

	Datasets	scSAGE _{NREC}	scSAGE _{NSE}	scSAGE _{NCON}	scSAGE _{NZINB}	scSAGE
NMI	Qs_Diaphragm	0.6962	0.854	0.9604	0.9687	0.9773
	Qs_Limb_Muscle	0.5828	0.9651	0.9538	0.9753	0.9753
	Qs_Lung	0.6851	0.8168	0.8337	0.8342	0.8342
	Muraro	0.6448	0.8885	0.8828	0.8799	0.8907
	Qx_Limb_Muscle	0.6771	0.9438	0.9855	0.9799	0.9809
	Young	0.4145	0.8052	0.7802	0.8014	0.8069
	Qx_Trachea	0.5684	0.7489	0.8658	0.9006	0.9012
	Qs_Trachea	0.6787	0.6925	0.9027	0.8994	0.8994
	Qx_Bladder	0.8754	0.9934	0.9786	0.9786	0.9809
	Klein	0.4373	0.6544	0.8151	0.8196	0.8199
ARI	Qs_Diaphragm	0.7673	0.7965	0.9799	0.9845	0.9893
	Qs_Limb_Muscle	0.4342	0.982	0.9713	0.9875	0.9875
	Qs_Lung	0.453	0.7096	0.7964	0.7972	0.7972
	Muraro	0.5411	0.924	0.9231	0.9195	0.9282
	Qx_Limb_Muscle	0.649	0.9553	0.9948	0.9925	0.9929
	Young	0.2099	0.6913	0.6743	0.687	0.6929
	Qx_Trachea	0.5839	0.8525	0.9617	0.9619	0.9688
	Qs_Trachea	0.8231	0.8333	0.939	0.9368	0.9368
	Qx_Bladder	0.94	0.9976	0.9878	0.9878	0.9893
	Klein	0.321	0.6433	0.8221	0.8208	0.8212

Table S9. A comprehensive robustness analysis is carried out on sparse data with sparsity levels set at 10%, 30%, and 50%.

		Datasets	scSAGE	scTAG	scDEFR	scGCOT	scDFC	scNAME	con-sc	scCCL	scGCL
Dropout_0.10	NMI	Qs_Diaphragm	0.9716	0.9398	0.9166	0.8885	0.9327	0.9359	0.5075	0.9338	0.9630
		Qs_Limb_Muscle	0.9519	0.8961	0.9492	0.9438	0.9116	0.9481	0.5024	0.9107	0.9125
		Qs_Lung	0.8169	0.7957	0.7938	0.8069	0.7585	0.7835	0.4606	0.7773	0.7344
		Muraro	0.8797	0.7783	0.8617	0.8079	0.5400	0.8140	0.4487	0.8747	0.8634
	ARI	Qs_Diaphragm	0.9863	0.9640	0.9328	0.9175	0.9613	0.9610	0.4031	0.9773	0.9815
		Qs_Limb_Muscle	0.9770	0.9081	0.9715	0.9631	0.9521	0.9764	0.3667	0.9538	0.9133
		Qs_Lung	0.7849	0.7464	0.6598	0.7810	0.74505	0.6204	0.2253	0.7429	0.4737
		Muraro	0.9184	0.7448	0.8580	0.8224	0.3501	0.7402	0.2850	0.9171	0.8475
Dropout_0.30	NMI	Qs_Diaphragm	0.9729	0.8392	0.8681	0.919	0.9299	0.9219	0.4239	0.9440	0.9636
		Qs_Limb_Muscle	0.9479	0.8894	0.9462	0.9246	0.9119	0.8549	0.4085	0.9174	0.9284
		Qs_Lung	0.8048	0.7435	0.7400	0.7281	0.7323	0.7876	0.4920	0.7616	0.7474
		Muraro	0.8838	0.7562	0.8010	0.8014	0.5755	0.8590	0.5150	0.8798	0.8543
	ARI	Qs_Diaphragm	0.9863	0.8566	0.9115	0.9516	0.9601	0.9487	0.2115	0.9763	0.9853
		Qs_Limb_Muscle	0.9705	0.9276	0.9615	0.9503	0.9547	0.9647	0.3275	0.9560	0.9114
		Qs_Lung	0.7371	0.5325	0.6934	0.5319	0.7226	0.6345	0.2498	0.7257	0.5032
		Muraro	0.9246	0.7828	0.6745	0.8480	0.3496	0.8999	0.3367	0.9200	0.8365
Dropout_0.50	NMI	Qs_Diaphragm	0.9558	0.8277	0.8356	0.9157	0.926	0.9092	0.3769	0.9092	0.9521
		Qs_Limb_Muscle	0.9182	0.7975	0.9032	0.8125	0.7879	0.8316	0.5211	0.8789	0.8911
		Qs_Lung	0.8243	0.7408	0.7410	0.7716	0.76679	0.8093	0.5106	0.6983	0.7269
		Muraro	0.8757	0.6763	0.7162	0.7279	0.7069	0.7620	0.4524	0.8679	0.8199
	ARI	Qs_Diaphragm	0.9772	0.8546	0.8666	0.9455	0.9601	0.9421	0.2942	0.9537	0.9758
		Qs_Limb_Muscle	0.9562	0.8132	0.9244	0.8168	0.6526	0.6711	0.3493	0.9231	0.8205
		Qs_Lung	0.7504	0.5568	0.7013	0.6701	0.7451	0.7371	0.2712	0.6948	0.4855
		Muraro	0.9204	0.6321	0.5496	0.7648	0.6656	0.6586	0.2598	0.9163	0.7348

Table S10. A comprehensive robustness analysis is carried out on noise data with noise levels set at 10%, 20%, and 40%.

			scSAGE	scTAG	scDEFR	scGCOT	scDFC	scNAME	con-sc	scCCL	scGCL
Noise_0.10	NMI	Qs_Diaphragm	0.9682	0.2933	0.8793	0.4478	0.9361	0.4161	0.4549	0.3090	0.7094
		Qs_Limb_Muscle	0.9105	0.4968	0.8954	0.4790	0.7855	0.7440	0.5339	0.6104	0.9041
		Qs_Lung	0.8410	0.5706	0.7814	0.6099	0.7461	0.7035	0.4755	0.5604	0.7455
		Muraro	0.8420	0.6283	0.8298	0.7783	0.5399	0.7899	0.4689	0.8258	0.8360
	ARI	Qs_Diaphragm	0.9836	0.2277	0.8992	0.2394	0.9637	0.2590	0.3484	0.1747	0.5545
		Qs_Limb_Muscle	0.9548	0.4462	0.9119	0.3073	0.6323	0.5757	0.4145	0.5233	0.9270
		Qs_Lung	0.8343	0.3092	0.6792	0.4262	0.7296	0.4264	0.2724	0.3483	0.5311
		Muraro	0.8854	0.5711	0.7893	0.7755	0.3534	0.7182	0.2664	0.7001	0.7465
Noise_0.20	NMI	Qs_Diaphragm	0.9638	0.268	0.8762	0.5009	0.9357	0.4819	0.4598	0.2779	0.7050
		Qs_Limb_Muscle	0.9146	0.3964	0.9003	0.4161	0.7807	0.7422	0.5329	0.6142	0.9008
		Qs_Lung	0.7832	0.5264	0.7821	0.6336	0.7313	0.7073	0.4800	0.5592	0.7725
		Muraro	0.8438	0.7288	0.8290	0.7743	0.5229	0.8067	0.4611	0.8362	0.8231
	ARI	Qs_Diaphragm	0.9826	0.2256	0.8922	0.2996	0.9653	0.3206	0.3529	0.1485	0.5424
		Qs_Limb_Muscle	0.9479	0.3054	0.9325	0.2162	0.6315	0.5564	0.4005	0.4991	0.9076
		Qs_Lung	0.7266	0.2498	0.6764	0.483	0.7237	0.4716	0.2465	0.3428	0.5876
		Muraro	0.8883	0.7093	0.8739	0.7918	0.3532	0.6872	0.2575	0.7200	0.7069
Noise_0.40	NMI	Qs_Diaphragm	0.9724	0.1792	0.8790	0.4185	0.9405	0.3792	0.4810	0.3406	0.9237
		Qs_Limb_Muscle	0.9299	0.4520	0.9170	0.4839	0.7831	0.6124	0.5417	0.1697	0.9122
		Qs_Lung	0.7918	0.5919	0.7818	0.6134	0.6865	0.6912	0.4693	0.5323	0.7464
		Muraro	0.8624	0.7731	0.8508	0.7769	0.5188	0.8136	0.4987	0.8195	0.8254
	ARI	Qs_Diaphragm	0.9854	0.0816	0.8951	0.2151	0.9681	0.2346	0.3631	0.1913	0.9428
		Qs_Limb_Muscle	0.9402	0.4299	0.9301	0.3580	0.6345	0.4128	0.4079	0.2013	0.9176
		Qs_Lung	0.8057	0.3068	0.6781	0.4515	0.6293	0.5185	0.2433	0.3282	0.5345
		Muraro	0.9014	0.7946	0.8642	0.7745	0.342	0.7422	0.3177	0.7106	0.7123

Table S11. The assignments of cell types to predicted clusters for PDAC dataset.

Cell Type	Cell Cluster	Overlap Genes number	Overlap ratio	Unique gene in cell type	Unique genes in predicted cluster
Monocyte	0	45	90%	SLC11A1 AIF1 LYZ SRGN COTL1 IGKC IGLC2 IGLC3	FYB ABTB1 SMCHD1 FAM129A LYN IRF8 LMAN1 FAM46C
B cell/Plasma	1	40	80%	IGHA1 IGHG1 JCHAIN IGHA2 IGHG4 IGHG3 IGHM	CRELD2 PPAPDC1B POU2AF1 HIFX KRTCAP2 ST6GAL1 TMEM59
Epithelial	2	45	90%	SPINK1 MMP7 GP2 PHLDA2 LAMB3	DSG2 MACROD2 MDK MUCL3 NQO1
Fibroblast	3	50	100%	NONE	NONE
Mast cell	4	48	96%	CALB2 LMNA	ALAS1 ARHGEF6
Macrophage	5	48	96%	SLCO2B1 TREM2	MS4A6A HLA-DMA
T cell	6	50	100%	NONE	NONE
Endothelial	7	47	94%	FABP5 FABP4 IFITM3	MT1A LIFR IGFBP4

Supplementary Pictures

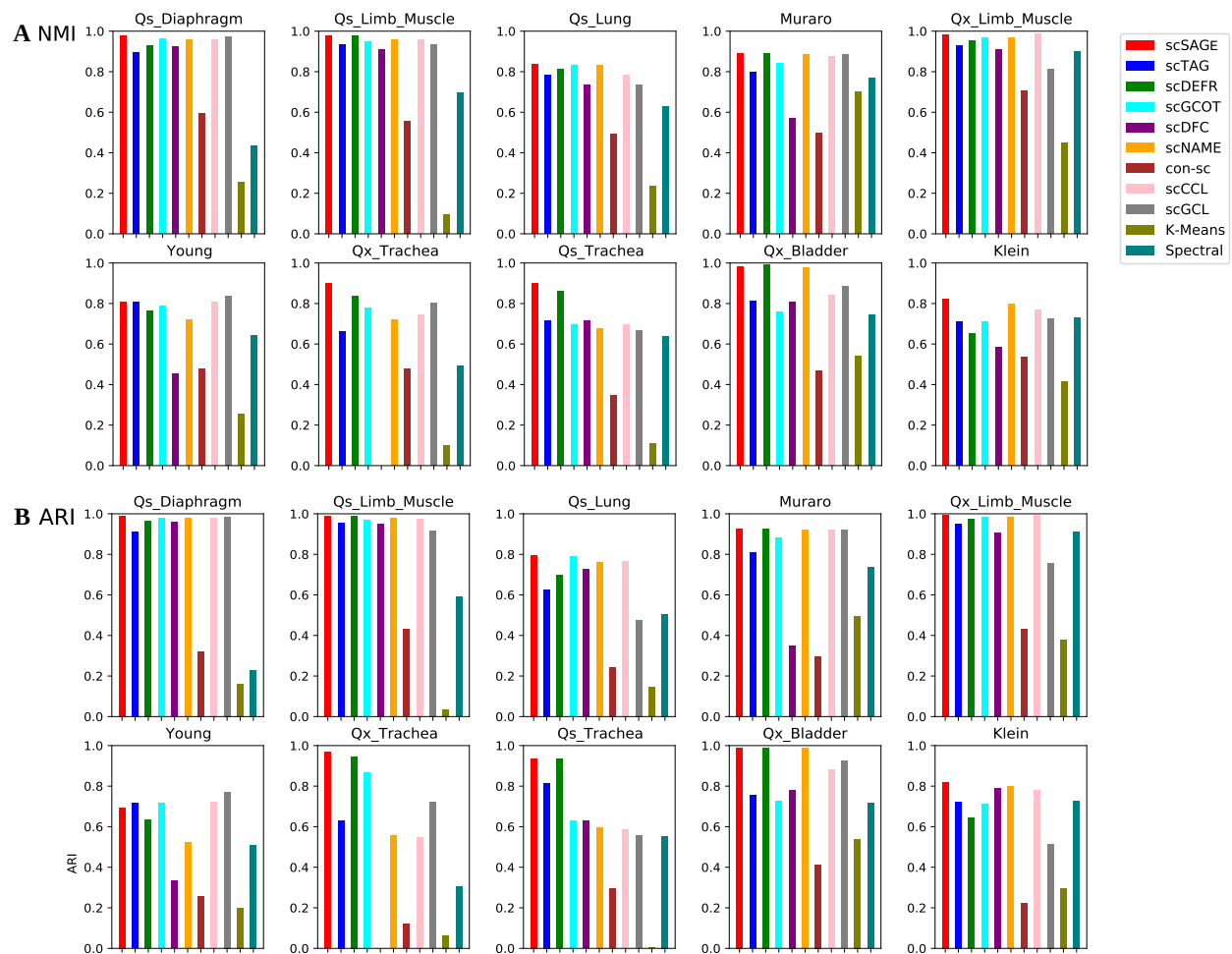


Figure S3. Clustering performance of scSAGE, scTAG, scDEFER, scGCOT, scDFC, scNAME, con-sc, scCCL, scGCL, K-Means and Spectral on ten scRNA-seq datasets measured by (A) NMI and (B) ARI.

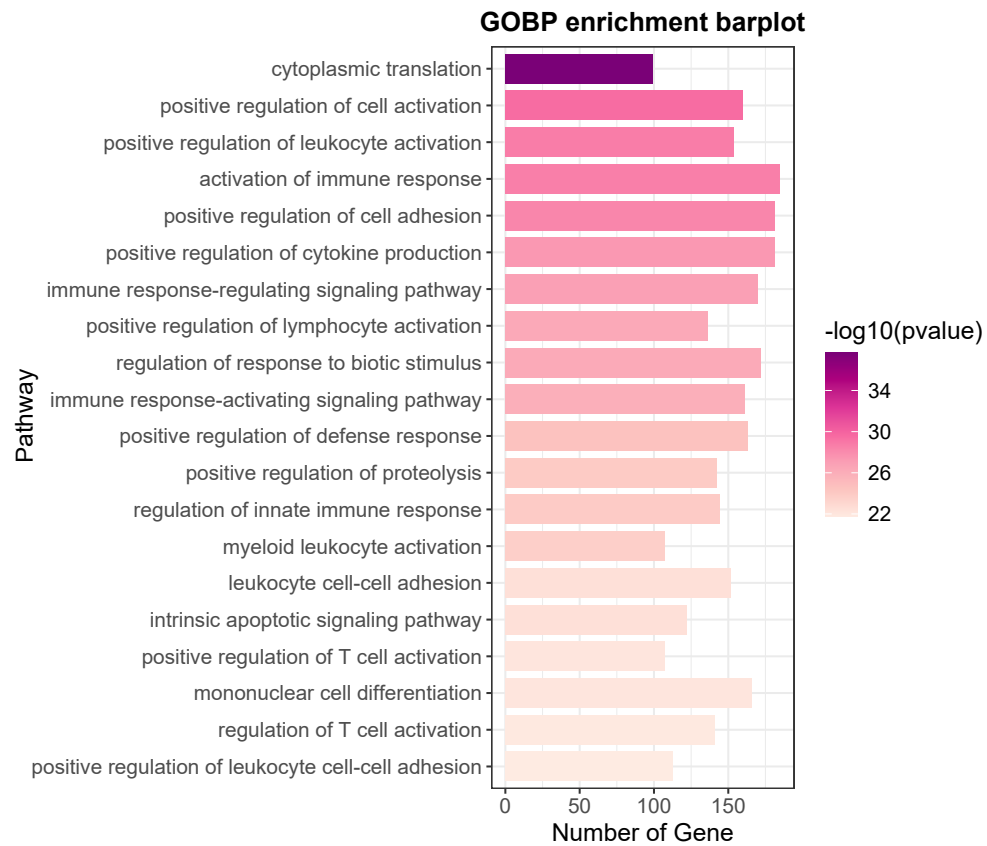


Figure S4. GOBP enrichment analysis for cluster 0.

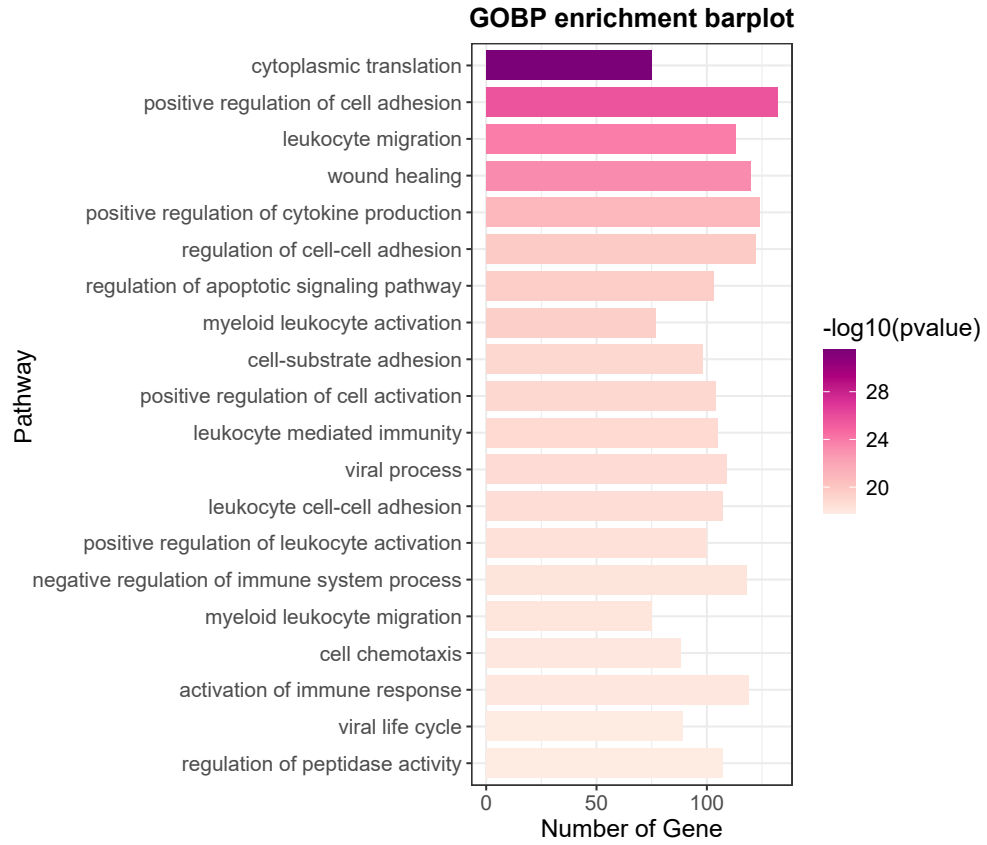


Figure S5. GOBP enrichment analysis for cluster 1.

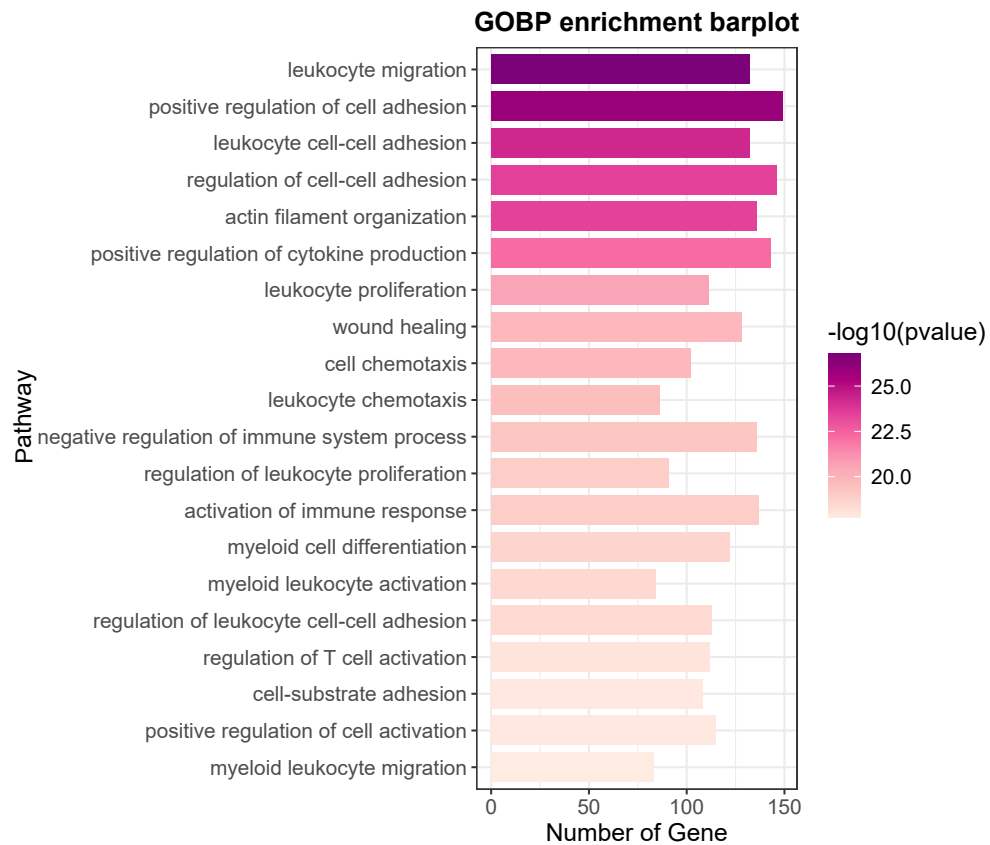


Figure S6. GOBP enrichment analysis for cluster 2.

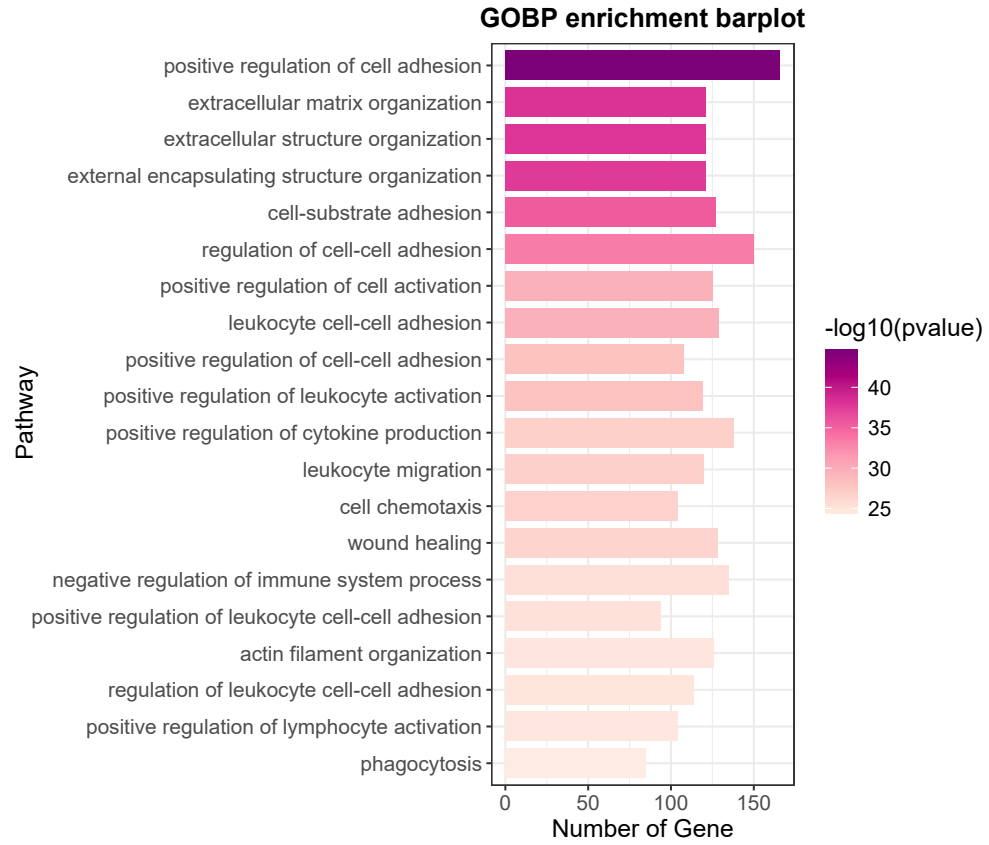


Figure S7. GOBP enrichment analysis for cluster 3.

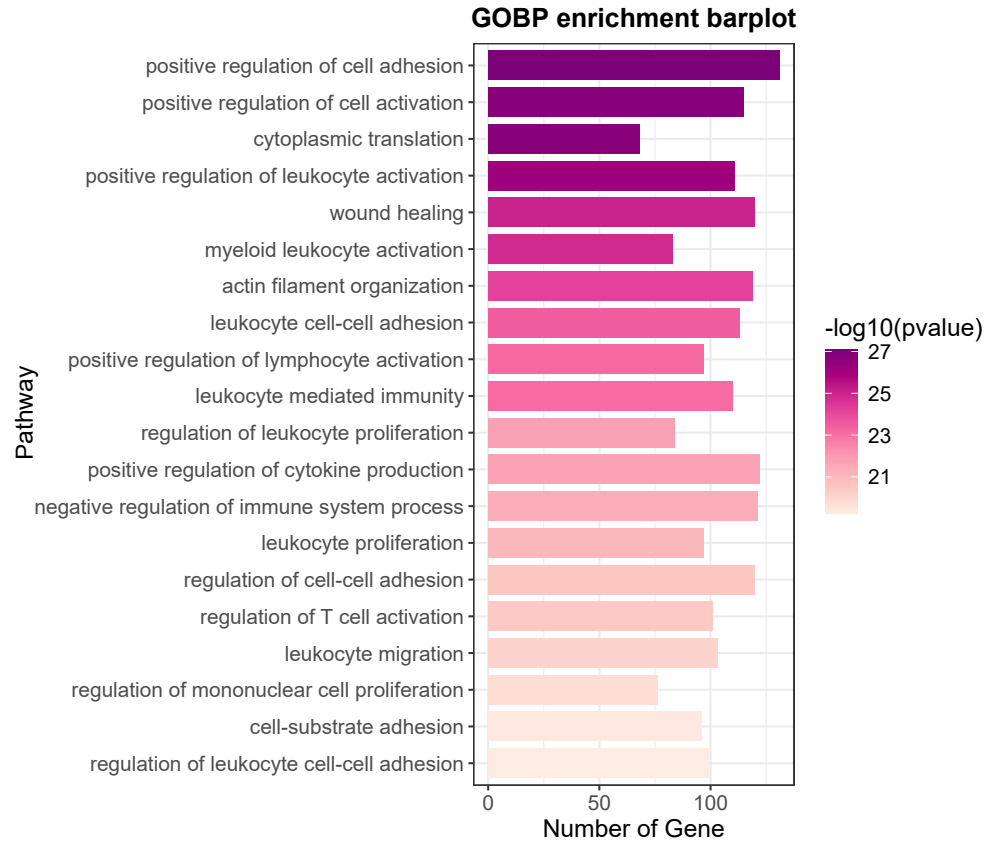


Figure S8. GOBP enrichment analysis for cluster 4.

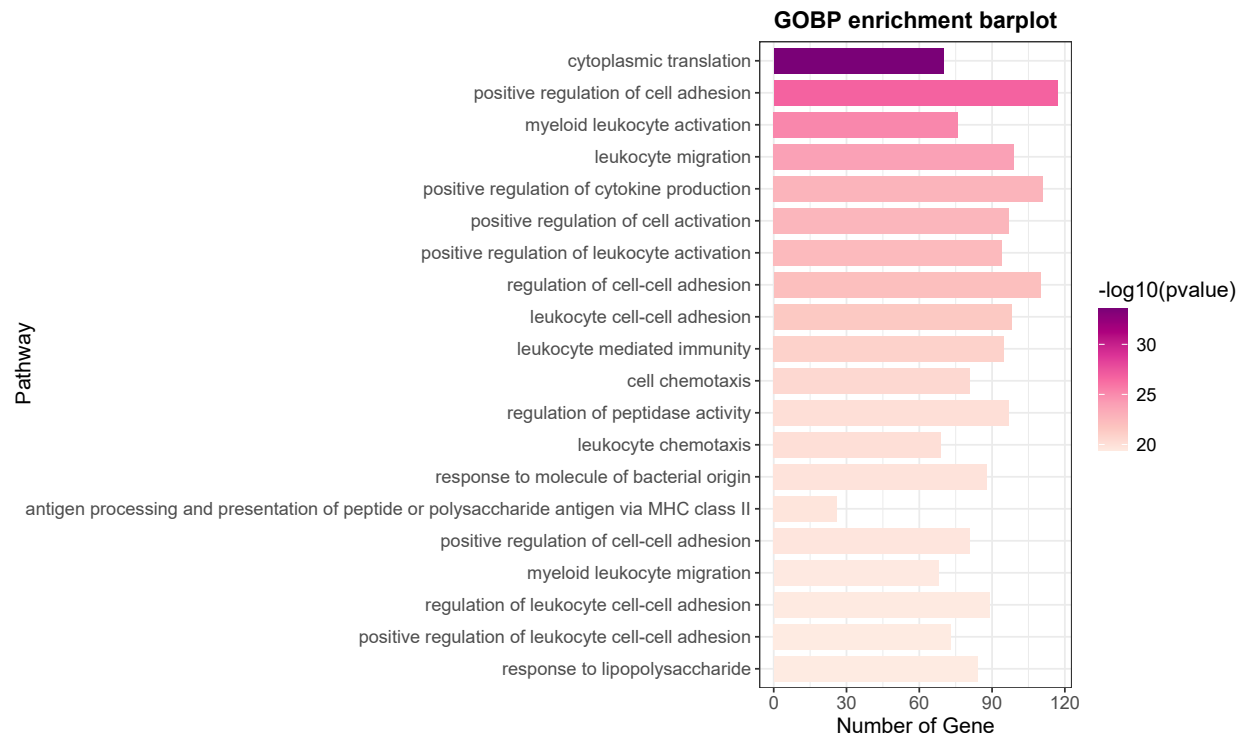


Figure S9. GOBP enrichment analysis for cluster 5.

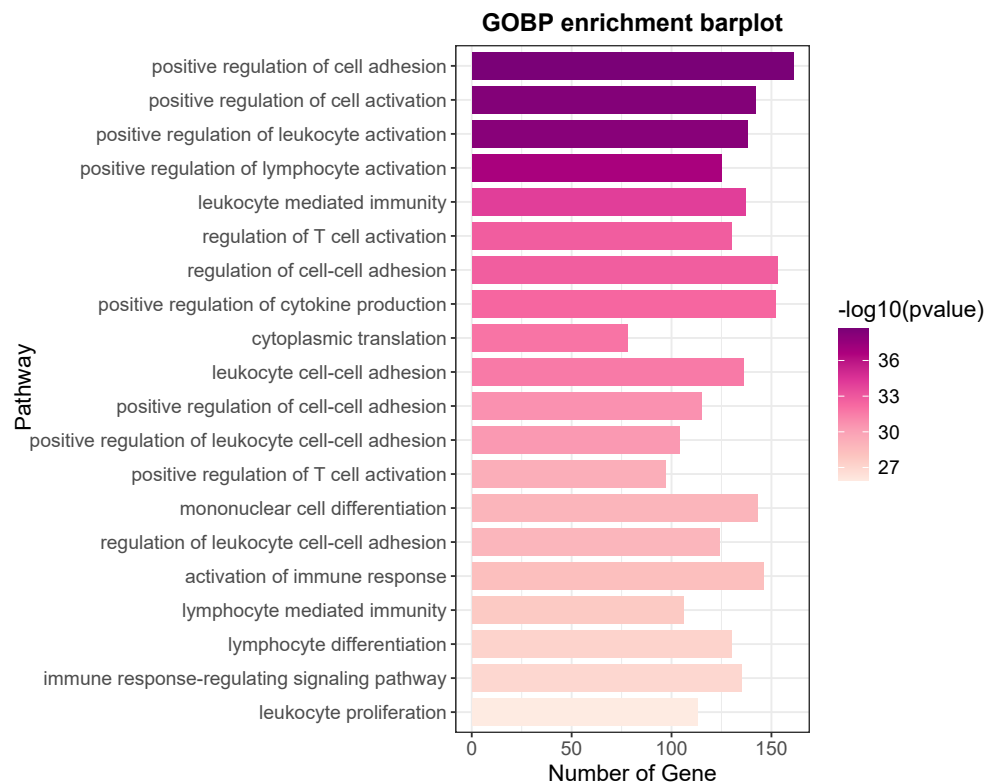


Figure S10. GOBP enrichment analysis for cluster 6.

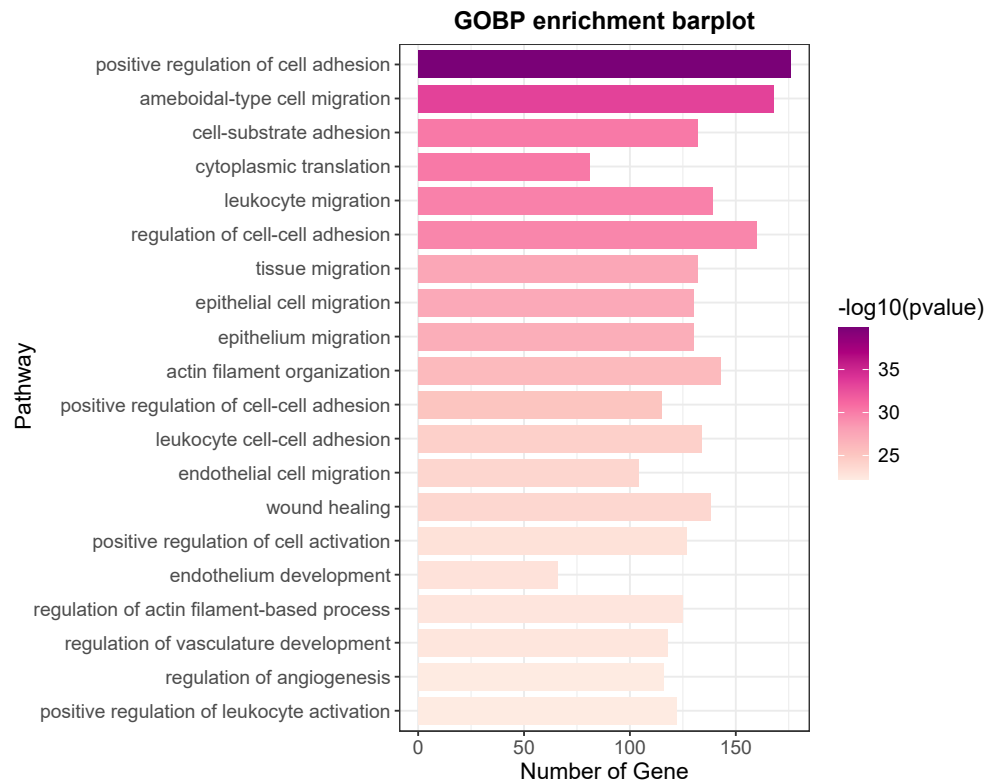


Figure S11. GOBP enrichment analysis for cluster 7.

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